

PRODUCTION SCOPE & MILESTONE DELIVERY SCHEDULE

Live Cockpit: <https://rag-agentic-studio.vercel.app>

MILESTONE	ARCHITECTURE & PRODUCTION ENGINEERING DELIVERABLES	TIMELINE	SHARE	ALLOCATION
Phase 0	Interactive Agentic RAG Simulator & Performance Cockpit (Live) Living demo: Operational hybrid semantic vector search, dual-provider auto-failover, NIST AI RMF governance dashboard, real-time citation matching, and multi-agent consensus workflow engine.	Live Now	Included	\$0.00 (Live)
Milestone 1	Discovery, Core RAG Document Ingestion & Chunking Pipeline Audit existing codebase, establish Dockerized FastAPI environment, implement document parser, set up semantic-aware chunking (with layout analysis/OCR matching), and store chunks on Supabase PgVector.	Week 1	20%	15 Hrs (\$600)
Milestone 2	Vector Search Indexing, Hybrid Retrieval & Cross-Encoder Reranking Configure HNSW index weights, integrate dense vector embeddings alongside sparse BM25 lexical search, deploy a lightweight cross-encoder reranker to isolate top 3 highly relevant snippets, and write unit tests.	Week 2	20%	15 Hrs (\$600)
Milestone 3	Multi-Agent Workflow Development, Consensus Checks & Orchestration Build Python-based AI agents with structured tasks (parsing, scoring, drafting), implement parallel execution loops, insert automated validator consensus checking, and implement state persistence.	Week 3	20%	20 Hrs (\$800)
Milestone 4	React / Next.js Interactive Frontend Features & API Integration Create high-performance interactive UI modules in React/Next.js, integrate with backend FastAPI endpoints, implement responsive charts, render real-time logs, and set up fluid smooth-scroll navigation.	Week 4	20%	15 Hrs (\$600)
Milestone 5	Multi-Provider AI Fallback, End-to-End Testing & Production Handover Deploy sub-500ms automated circuit breakers (OpenAI to Gemini), insert OWASP LLM01 anti-injection filters, configure Docker Compose/AWS deployment structures, and supply zero lock-in handover guides.	Week 5	20%	15 Hrs (\$600)
TOTAL AI FULL STACK ENGINEER SCOPE (TURNKEY PYTHON RAG & AGENTS)		5 Sprints	100%	80 Hrs (\$3,200)

PHASED AI SYSTEM DELIVERY ROADMAP

Phase 0: Interactive Architecture Cockpit (Delivered)

\$0.00 (Live Ahead of Bid)

Milestone 1: Code Discovery & Ingestion Pipeline

15 Hrs (Net Day 7)

Milestone 2: Vector DB Indexing & Hybrid Retrieval

15 Hrs (Net Day 14)

Milestone 3: Python Multi-Agent Core Workflows

20 Hrs (Net Day 21)

Milestone 4: Next.js Responsive Frontend UI & Charts

15 Hrs (Net Day 28)

Milestone 5: AI Fallback, LLM Firewalls & Handover

15 Hrs (Net Day 35)

PRODUCTION AI & SOFTWARE DELIVERY GUARDRAILS

✓ Hybrid RAG Precision: Dense vector + BM25 sparse search ensures highly accurate keyword and semantic matches.

✓ AI Fallback Reliability: Multi-provider automated gateway guarantees sub-500ms failover from OpenAI to Gemini.

✓ Zero Lock-in Blueprinting: All configurations, Python services, and API structures are standard-compliant and portable.

✓ Clean Docker Scaffolding: Everything deploys in unified, production-ready Docker containers with compose files.

✓ Daily Async Standups: Structured updates on shipped features, next PRs, and test pass rates. Zero hand-holding.

CALIBRATED HOURLY RATE

\$40.00/hr capped at 80 total engineering hours. Perfect alignment with client rate preferences (\$22-\$40/hr).

CLOUD NATIVE OPTIMIZATION

FastAPI asynchronously multiplexes requests to PostgreSQL/PgVector pools, cutting compute costs by 50%.

100% CODE OWNERSHIP

All code, PRs, Terraform modules, and Docker configs committed directly to client's Git repositories.

DAILY ASYNC STANDUPS

Daily notes on shipped work, upcoming PRs, blockers, and test status. Zero hand-holding required.

FORMAL AUTHORIZATION & AI ENGINEERING ACCEPTANCE

BINDING UPON CONTRACT ACTIVATION VIA UPWORK HOURLY OFFER / MILESTONE SCHEDULE

AUTHORIZED ARCHITECT: BARAKAHSOFT LLC (WYOMING, USA)

Signatory: **Shakil Ahmed** • Principal Systems Architect & Former Lead Engineer at Legiit

Shakil Ahmed

19 Sep 2026

AUTHORIZED ARCHITECT SIGNATURE

DATE

AUTHORIZED CLIENT: ENTERPRISE SAAS PLATFORM

Signatory: **Founder / CTO** • Authorized SaaS Client Representative

[Accepted via Upwork Contract Offer / Sign-off]

___ / ___ / 2026

AUTHORIZED CLIENT SIGNATURE

DATE